

Final Incident Report

Project	HOME SOC LAB
Scenario	Windows Endpoint Compromise Simulation
Incident Type	RDP-Based Intrusion
Severity	High
Status	Closed
Date	24 August 2026

1. Executive Summary

A controlled intrusion scenario was conducted against the Windows 11 SOC endpoint from the Kali Linux attack host.

The activity began with repeated RDP authentication failures against the Windows host. Following the failed authentication attempts, a successful Remote Desktop session was established using the WIN11\ vboxuser account.

After authentication, activity continued within the Windows session. PowerShell was executed, followed by local account discovery using the net.exe utility. Persistence was subsequently established through creation of a Windows Scheduled Task.

The activity ultimately progressed to Defense Evasion, where the Windows Application event log was cleared.

The complete sequence was visible through Windows Security and Sysmon telemetry and was detected by Wazuh at multiple stages.

The observed attack progression was:

RDP Brute Force → Successful RDP → PowerShell Execution → Account Discovery → Persistence → Defense Evasion

The exercise successfully demonstrated the ability to detect and investigate a multi-stage Windows intrusion using endpoint telemetry and SIEM correlation.

2. Incident Overview

Category	Details
Incident	Simulated Windows Endpoint Intrusion
Severity	High
Status	Closed
Source	Kali Linux

Category	Details
Source IP	192.168.56.105
Target	Windows 11
Target IP	192.168.56.106
Account	WIN11\vbouser
Remote Service	RDP
SIEM	Wazuh
Endpoint Telemetry	Windows Security / Sysmon
Network	Isolated Host-Only Lab

3. Incident Description

The simulated intrusion targeted the Windows 11 SOC endpoint through Remote Desktop Protocol.

Multiple authentication attempts were first generated against the endpoint. Windows recorded failed authentication events and Wazuh detected the repeated failures.

The attacker subsequently authenticated successfully through RDP. Windows recorded a successful remote interactive logon using **Event ID 4624 with Logon Type 10**.

After gaining access, PowerShell execution was observed. Sysmon recorded process creation activity and identified a PowerShell process spawning another PowerShell process.

The authenticated session then performed local account discovery using:

net.exe localgroup administrators

The activity progressed to persistence through creation of the scheduled task:

\SOC-Lab-persistence-Test

The scheduled task contained a logon trigger and was successfully recorded by Windows Security.

The final stage involved clearing the Windows Application event log. Windows generated **Event ID 104**, and Wazuh detected the activity as a Windows log-clearing event.

4. Attack Timeline

Phase Activity		Windows / Sysmon Evidence Wazuh	
1	RDP authentication attempts	Event ID 4625	60122 / 60204
2	Successful RDP authentication	Event ID 4624 / Logon Type 10	Detected
3	PowerShell execution	Sysmon Event ID 1	92027
4	Account discovery	Sysmon Event ID 1	92033
5	Scheduled task creation	Event ID 4698	60228
6	Application log clearing	Event ID 104	63104

5. Initial Access

The attack originated from the Kali Linux host:

Source: 192.168.56.105

Target: 192.168.56.106

Service: RDP / TCP 3389

Multiple failed authentication attempts were generated.

Windows Security recorded **Event ID 4625** for the failed attempts.

Wazuh detected the authentication failures and generated alerts for both individual failures and multiple logon failures.

This established the initial attack activity against the Windows endpoint.

MITRE ATT&CK:

T1110 — Brute Force

6. Successful Remote Access

Following the failed authentication activity, the attacker successfully authenticated through Remote Desktop.

Windows Security recorded:

Event ID: 4624

Logon Type: 10

The successful authentication established an interactive RDP session under:

WIN11\wboxuser

This event represents the transition from attempted access to authenticated access on the Windows endpoint.

MITRE ATT&CK:

T1021.001 — Remote Services: Remote Desktop Protocol

7. Post-Authentication Execution

Following successful RDP authentication, PowerShell activity was observed.

Sysmon Event ID 1 recorded:

Image:

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

The parent process was also PowerShell:

ParentImage:

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

The activity occurred under:

WIN11\ vboxuser

Wazuh generated **Rule 92027, Level 4**.

The process relationship demonstrates execution occurring within the authenticated RDP session.

MITRE ATT&CK:

T1059.001 — Command and Scripting Interpreter: PowerShell

8. Discovery

The authenticated session subsequently executed:

"C:\Windows\system32\net.exe" localgroup administrators

The command queried membership of the local Administrators group.

Sysmon recorded the process creation and identified PowerShell as the parent process.

Wazuh generated **Rule 92033**, identifying discovery activity associated with PowerShell execution.

This activity demonstrates local account and privilege-group discovery following successful access to the endpoint.

MITRE ATT&CK:

T1087 — Account Discovery

9. Persistence

Persistence was established through creation of the Windows Scheduled Task: \SOC-Lab-persistence-Test

Windows Security generated: Event ID: 4698

The task contained a logon trigger: LogonTrigger

and executed: cmd.exe /c echo SOC-LAB-PERSISTENCE-TEST

The task was associated with: WIN11\vbouser

Wazuh generated:

Rule: 60228

Level: 4

Description: A scheduled task was created

The scheduled task demonstrated the use of Windows Task Scheduler as a persistence mechanism.

MITRE ATT&CK:

T1053 — Scheduled Task/Job

10. Defense Evasion

The final stage involved clearing the Windows Application event log.

Windows generated: Event ID: 104

With the message: The Application log file was cleared.

The affected channel was: Application

The event was associated with: WIN11\vbouser

Wazuh generated:

Rule: 63104

Level: 5

Description: A Windows log file was cleared

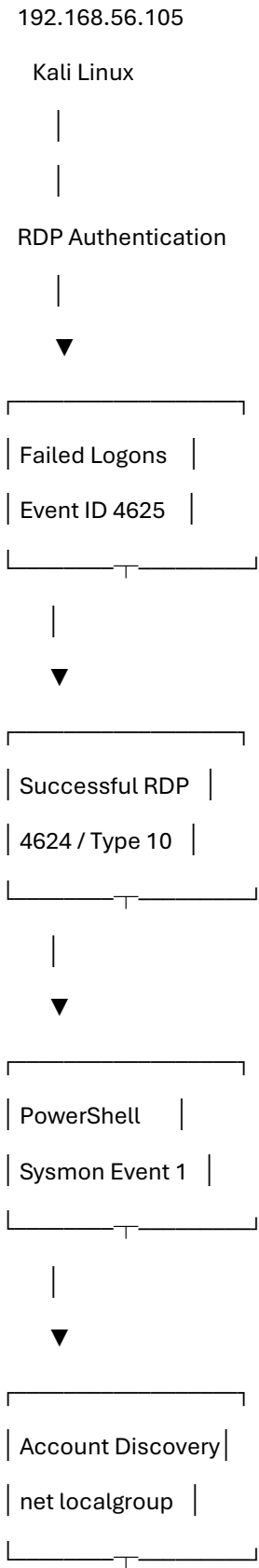
The activity represents an attempt to remove or reduce available forensic evidence from the endpoint.

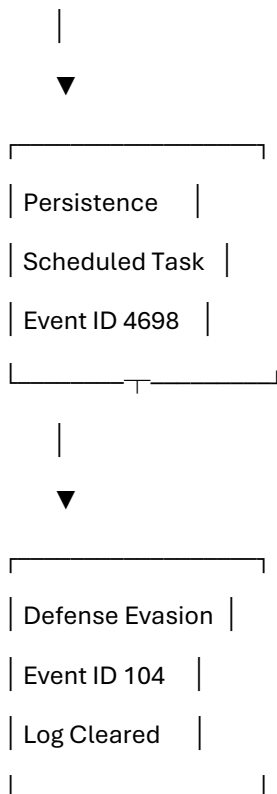
MITRE ATT&CK:

T1070 — Indicator Removal

11. Correlated Attack Path

The complete incident can be reconstructed as:





12. MITRE ATT&CK Mapping

Stage	Technique	ID
Initial Access	Brute Force	T1110
Remote Access	Remote Services: RDP	T1021.001
Execution	PowerShell	T1059.001
Discovery	Account Discovery	T1087
Persistence	Scheduled Task/Job	T1053
Defense Evasion	Indicator Removal	T1070

13. Key Indicators

Source

192.168.56.105

Target

192.168.56.106

Account

WIN11\ vboxuser

Remote Service

RDP / TCP 3389

Discovery Command

net.exe localgroup administrators

Scheduled Task

\SOC-Lab-persistence-Test

Cleared Log

Application

14. Detection Assessment

The incident demonstrated successful detection at every major stage.

Authentication attacks were detected through Windows Security events and Wazuh rules.

Successful RDP access was confirmed through Event ID 4624 and Logon Type 10.

PowerShell execution was visible through Sysmon process telemetry.

Discovery activity was detected through the net.exe process and Wazuh Rule 92033.

Persistence was detected through Windows Event ID 4698 and Wazuh Rule 60228.

Defense Evasion was detected through Windows Event ID 104 and Wazuh Rule 63104.

The combination of Windows Security, Sysmon, and Wazuh provided sufficient telemetry to reconstruct the simulated intrusion.

15. Incident Response

If this activity occurred without authorization in a production environment, the following actions would be recommended:

1. Terminate the suspicious RDP session.
2. Isolate the affected endpoint if compromise is confirmed.
3. Validate the originating IP address.
4. Identify and secure the affected account.
5. Review the complete PowerShell process tree.
6. Investigate discovery commands executed during the session.

7. Inspect the newly created scheduled task.
8. Preserve available Windows Security and Sysmon telemetry.
9. Investigate the event-log clearing activity.
10. Determine whether additional forensic evidence was removed.
11. Review other systems for related authentication attempts.
12. Reset credentials if unauthorized access is confirmed.

16. Evidences

All Evidences and the Screenshots are uploaded in the Individual Reports .

17. Final Incident Assessment

The HOME SOC LAB successfully demonstrated a complete simulated intrusion against a Windows endpoint.

The attacker progressed from repeated RDP authentication attempts to successful remote access and subsequently performed execution, discovery, persistence, and Defense Evasion activities within the authenticated session.

The correlated activity demonstrates the following attack progression:

Brute Force → Successful RDP → PowerShell Execution → Account Discovery → Scheduled Task Persistence → Indicator Removal

Wazuh successfully detected the major stages of the activity through Windows Security and Sysmon telemetry.

The incident demonstrates practical capability in:

- Windows endpoint monitoring
- RDP attack detection
- Process-tree analysis
- Discovery detection
- Persistence detection
- Defense Evasion detection
- MITRE ATT&CK mapping
- SIEM investigation
- Attack-chain reconstruction
- Incident response

Final Verdict:

SIMULATED WINDOWS ENDPOINT COMPROMISE — DETECTED AND CORRELATED SUCCESSFULLY

Incident Status: CLOSED

Classification: Controlled Home SOC Laboratory Exercise

No evidence from this exercise indicates compromise of the physical host machine.